

Finite balance of endpoint odds on a growing region

Asymptotic research note; independently reviewed

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Let $p_1, \dots, p_m : [K] \rightarrow [0, 1]$ be nondecreasing, with positive probability weights w_q , and suppose

$$\mathbb{E}_w \prod_{j \in S} p_j = \frac{1}{|S| + 1} \quad (S \subseteq [m]).$$

Write $A = \prod_j p_j$. On $A > 0$ set $r_i = (1 - p_i)/p_i$ and $\lambda = \sum_i r_i$. All constants below are absolute; equal row weights are not assumed.

Lemma 1 (Failure tilts with a variable number of failures). *Let $J \subseteq [m]$, $\ell = |J|$, $d = m - \ell > 0$, and put*

$$\nu_J(q) = (m + 1) \binom{m}{\ell} w_q \prod_{j \in J} (1 - p_j(q)) \prod_{i \notin J} p_i(q).$$

This is a probability measure. On its support all p_i , $i \notin J$, are positive. For $\varepsilon > 0$, let $G = \{\max_{i \notin J} r_i < \varepsilon\}$, and write $h_k = \binom{\ell + k}{\ell}$. Then, for $0 \leq k \leq d$,

$$h_k \left(1 - \frac{(\ell + k + 1)/\varepsilon + k}{d} \right) \leq \mathbb{E}_{\nu_J} [e_k(r_{-J}) \mathbf{1}_G] \leq h_k. \quad (1)$$

For every $i \notin J$,

$$\mathbb{E}_{\nu_J} r_i \leq \frac{\ell + 1}{d}. \quad (2)$$

Proof. Expansion of the mixed moments gives probability $1/((m + 1) \binom{m}{s})$ for every specified failure pattern of size s . Consequently the undivided generating polynomial

$$(m + 1) \binom{m}{\ell} \mathbb{E}_w \left[\prod_{j \in J} (1 - p_j) \prod_{i \notin J} (p_i + z(1 - p_i)) \right] = \sum_{k=0}^d h_k z^k. \quad (3)$$

If one specified factor $i \notin J$ is replaced by $1 - p_i$, its coefficient of z^k is

$$b_k = \frac{k+1}{d} \binom{\ell+k+1}{\ell} = \frac{\ell+k+1}{d} h_k \quad (0 \leq k < d), \quad (4)$$

with $b_d = 0$. Its constant coefficient proves (2): cancellation of p_i can additionally count nonnegative terms from zero entries, which explains the possible inequality.

For completeness apply the bad-prefix argument to the original ordered rows, including those having zero entries. The set where some $p_i = 0$ or $r_i \geq \varepsilon$, $i \notin J$, is a prefix. If it is nonempty, choose a column i witnessing badness at its last row. On the whole prefix,

$$p_i + z(1 - p_i) \leq (\varepsilon^{-1} + z)(1 - p_i)$$

coefficientwise. Thus its contribution to coefficient k of (3) is at most $\varepsilon^{-1}b_k + b_{k-1}$, where $b_{-1} = 0$. For $k < d$, (4) and $b_{k-1} = kh_k/d$ give the claimed loss. For $k = d$ the same asserted loss is only larger, so remains valid. Outside that prefix all the denominators are positive, and (3) becomes the ν_J expectation in (1). Positivity also proves its upper bound. $\square$

Theorem 1 (Uniform finite balance). *If $m \geq 256$, then every row q with $A(q) > 0$ and $\lambda(q) \leq m/128$ satisfies*

$$\max_{1 \leq i \leq m} r_i(q) \leq \frac{64(\lambda(q) + 1)}{m}. \quad (5)$$

In particular the maximum odds tends uniformly to zero on every region $\lambda = o(m)$, with the explicit rate in (5).

Proof. Fix such a row, put $L = \lambda(q)$, and fix a column i . Choose J avoiding i with $\ell = \lceil 4L \rceil + 4$ elements. Then $4 \leq \ell \leq m/16$, $d = m - \ell \geq 15m/16$, and $(\ell + 3)/d \leq 1/8$. Use the preceding lemma with $\varepsilon = 1$, and under ν_J define the nonnegative random variable $Z = \lambda_{-J} \mathbf{1}_G$. Its first moment obeys

$$\mathbb{E}Z \geq (\ell + 1) \left(1 - \frac{\ell + 3}{d} \right) \geq \frac{7}{8}(\ell + 1).$$

On G , $\sum_{i \notin J} r_i^2 \leq \lambda_{-J}$, whence

$$\mathbb{E}Z^2 \leq 2\mathbb{E}[e_2(r_{-J}) \mathbf{1}_G] + \mathbb{E}Z \leq (\ell + 1)(\ell + 2) + (\ell + 1) = (\ell + 1)(\ell + 3).$$

Since $L \leq \ell/4$, Cauchy–Schwarz gives

$$\nu_J(Z > L) \geq \frac{(\mathbb{E}Z - L)^2}{\mathbb{E}Z^2} \geq \frac{(5\ell/8)^2}{(5\ell/4)(7\ell/4)} = \frac{5}{28} > \frac{1}{8}.$$

Every row in this event has $\lambda_{-J} > L \geq \lambda_{-J}(q)$, and hence lies strictly before q . Monotonicity of r_i and (2) therefore yield

$$\frac{1}{8}r_i(q) \leq \mathbb{E}_{\nu_J} r_i \leq \frac{\ell+1}{d}.$$

Finally $\ell+1 \leq 4L+6$ and $d \geq 15m/16$, so

$$r_i(q) \leq \frac{128(4L+6)}{15m} \leq \frac{64(L+1)}{m}.$$

The choice of i was arbitrary. $\square$

Theorem 2 (Balance in the mean-deficit coordinate). *Put $s(q) = \sum_{j=1}^m (1 - p_j(q))$. If $m \geq 256$, every row with $s(q) \leq m/256$ has all $p_i(q) > 0$ and satisfies*

$$\max_i r_i(q) \leq \frac{128(s(q)+1)}{m}. \quad (6)$$

Thus this conclusion does not require an a priori bound on the odds, or even an a priori assumption that the row entries are positive.

Proof. Fix a target row q , write $s = s(q)$, and fix i . Choose $J \subseteq [m] \setminus \{i\}$ of size $\ell = \lceil 8s \rceil + 4$ that contains every $j \neq i$ with $p_j(q) \leq 1/2$. Such a choice is possible: there are at most $2s$ of these columns, and $4 \leq \ell \leq m/16$. Put $d = m - \ell$ and $H = [m] \setminus (J \cup \{i\})$. At the target row all columns in H are positive, and

$$\lambda_H(q) = \sum_{j \in H} r_j(q) \leq 2s \leq \ell/4. \quad (7)$$

Use the same probability measure ν_J as above, but generate additional failures only among H , retaining the success factor p_i . Its undivided generating polynomial has exact coefficients

$$c_k = \frac{d-k}{d} \binom{\ell+k}{\ell} \quad (0 \leq k \leq d-1). \quad (8)$$

This follows either by counting the $\binom{d-1}{k}$ possible additional failure patterns, or directly from (3). For a specified $j \in H$, its derivative coefficients are

$$b'_k = \frac{d-1-k}{d-1} b_k \leq b_k \quad (0 \leq k \leq d-2), \quad (9)$$

where b_k is (4).

Let $G = \{\max_{j \in H} r_j < 1\}$, including all zero entries among H in the complementary bad prefix. The same coefficientwise prefix argument, now using (8)–(9), gives

$$\mathbb{E}_{\nu_J}[\lambda_H \mathbf{1}_G] \geq (\ell + 1) \left(1 - \frac{\ell + 4}{d}\right) \geq \frac{7}{8}(\ell + 1).$$

The upper bounds for the first and second elementary symmetric moments are still h_1 and h_2 . Therefore, for $Z = \lambda_H \mathbf{1}_G$,

$$\mathbb{E}_{\nu_J} Z^2 \leq (\ell + 1)(\ell + 3).$$

The previous Cauchy–Schwarz estimate now shows $\nu_J(Z > 2s) \geq 1/8$. By (7) and monotonicity, these rows lie strictly before q . If $p_i(q) = 0$, p_i vanishes on that entire prefix, contradicting its positive ν_J -mass. Thus $p_i(q) > 0$. On this prefix $r_i \geq r_i(q)$, so (2) yields

$$r_i(q) \leq \frac{8(\ell + 1)}{d} \leq \frac{128(8s + 6)}{15m} \leq \frac{128(s + 1)}{m}.$$

This holds for every i . □

Corollary 1 (Original row mass near the endpoint). *For $m \geq 256$ and every $S \geq 0$,*

$$\sum_{q: s(q) \leq S} w_q \leq \frac{512(S + 1)}{m}. \quad (10)$$

Consequently, for every $c > 0$,

$$\mathbb{E}_w e^{-cs} \leq \frac{512(1 + c^{-1})}{m}. \quad (11)$$

For every integer $a \geq 1$,

$$\mathbb{E}_w [s^a e^{-cs}] \leq \frac{512a!}{m} \left(\frac{2a + 1}{c^{a+1}} + \frac{2}{c^a} \right). \quad (12)$$

Proof. If $S > m/256$, (10) follows from total mass one. Otherwise Theorem 2 implies, on $s \leq S$, that every product of k columns is at least $\exp(-128k(S + 1)/m)$. Choose $k = \lfloor m/(128(S + 1)) \rfloor$, which is at most m . The exact mixed moment $1/(k + 1)$ gives

$$\Pr_w(s \leq S) \leq \frac{e}{k + 1} \leq \frac{128e(S + 1)}{m} \leq \frac{512(S + 1)}{m}.$$

This also covers $k = 0$. If $F(S) = \Pr_w(s \leq S)$, Tonelli gives $\mathbb{E}e^{-cs} = \int_0^\infty ce^{-ct}F(t) dt$, proving (11), including a possible atom at zero. For $a \geq 1$, integration by parts and (10) give

$$\mathbb{E}[s^a e^{-cs}] \leq \frac{512}{m} \int_0^\infty (t+1)(at^{a-1} + ct^a)e^{-ct} dt,$$

which evaluates to (12). □

The estimate uses the full family of failure-pattern identities, with a number of conditioned failures proportional to λ . It does not follow merely from approximate moments of the single endpoint measure. By itself it gives no improved growth rate for K .